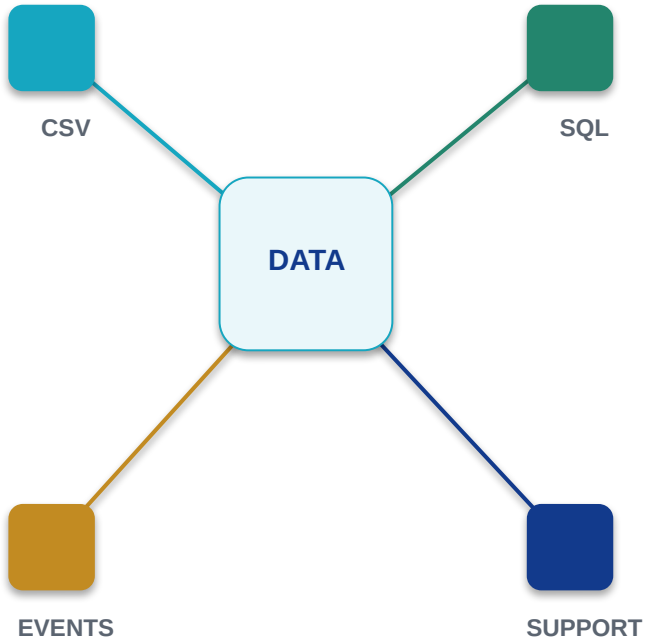
A3-PS019-TC279

PERSONAL-DATA DISCOVERY AND PRIVACY EVIDENCE INVENTORY



TraceLens

Discover personal data. Trace where it appears. Identify missing evidence.

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AIORI Hackathon 3 · Chitkara University

October 14, 2026

Abstract & Problem Description

Abstract: DATAQUEST scans structured sources and creates an evidence-backed inventory of potential personal-data fields, possible cross-source overlaps, and missing purpose/retention information.

Problem Description

- Personal data can be stored across CSV files and SQL tables.
- The same kind of data may appear under different field names.
- Purpose and retention information may be missing or difficult to locate.
- Manual searches make cross-source discovery hard to repeat.
- The goal is a clear, reviewable inventory—not an automatic legal decision.

Background

The prototype follows a simple, reviewable workflow.

- Input sources: multiple CSV files and a SQLite database.
- Profiling checks field names, data types, and sample values.
- Configurable rules identify potential matches to five configured prototype classes.
- Footprint matching groups possible overlaps across sources.
- Metadata checks show whether purpose and retention evidence is present.
- Human review and a manually labelled sample help assess the results.

Literature & RFC Review

Relevant guidance used as context for the design.

- NIST Privacy Framework 1.0 (2020): supports identifying and managing privacy risks [1].
- GDPR Article 5: includes purpose limitation, data minimisation, and storage limitation [2].
- GDPR Article 30: describes records of processing activities for applicable organisations [2].
- Design takeaway: record where a potential field appears and whether purpose/retention evidence exists.
- Boundary: the prototype provides technical evidence; it does not determine legal compliance.

Sources [1] and [2] are listed in the bibliography.

Proposed Solution

DATAQUEST is a local web prototype for structured data discovery.

- Load multiple CSV files and a SQLite database.
- Scan fields using configurable aliases, patterns, and evidence signals.
- Show potential personal-data findings and their source locations.
- Group possible repeated values across sources into a footprint.
- Map declared purpose and retention information and flag missing evidence.
- Let a human accept, reject, or revisit uncertain findings; export reports.

JUDGE DEMO: guardian_phone in admissions.csv → inspect its possible footprint across admissions.csv, events.csv, and support.db → check metadata → review the finding.

Optimization Proposed

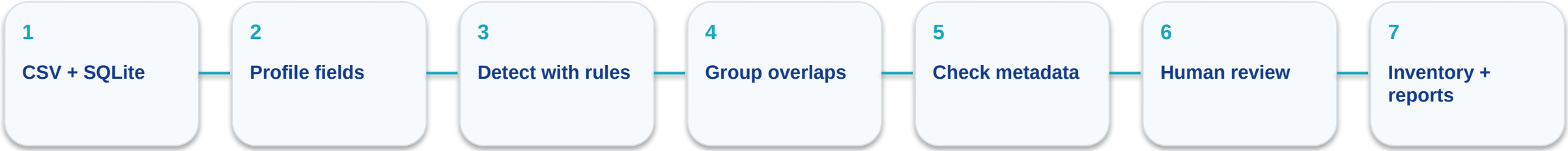
Before-and-after comparison of the workflow.

AREA	BEFORE	AFTER
Discovery	Search files/tables separately	Scan CSV + SQLite in one workflow
Evidence	Rationale is scattered	Show field location and detection evidence
Footprint	Repeated values are hard to spot	Group possible overlaps across sources
Metadata	Gaps can be missed	Highlight missing purpose/retention evidence
Review	Uncertainty handled informally	Record review decisions and notes
Evaluation	Hard to measure results	Calculate per-class precision and recall

Evaluation uses 19 manually labelled fields across five configured classes: useful for a small check, not proof of general accuracy.

Solution Architecture & Design

Flowchart of the main processing steps.



Supporting inputs: detection rules · purpose/retention declarations · manually labelled evaluation set

Detection strength is evidence strength, not a calibrated probability.

Data Flow Diagram

Overview of the inputs, processing, and outputs.



Metadata declarations → purpose / retention checks

Manual labels → precision / recall evaluation

Files are processed locally; use synthetic demo data, not real sensitive records.

Deployment Timeline

Milestones for a reproducible demo and validation.

1

Prepare

Confirm required classes
Use synthetic data

2

Run

Start backend/frontend
Check API health

3

Scan

Review inventory
footprint + metadata

4

Validate

Run tests
state limits honestly

5

Submit

Upload PDF, code
and demo link

Gate before presentation: verify the full workflow end-to-end on the submission machine.

Bibliography

Sources referenced in the presentation.

- [1] National Institute of Standards and Technology. NIST Privacy Framework, Version 1.0, 2020. https://www.nist.gov/privacy-framework

- [2] European Parliament and Council. Regulation (EU) 2016/679 (GDPR), Articles 5 and 30. https://eur-lex.europa.eu/eli/reg/2016/679/oj

- [3] AIORI-3 Hackathon. PS-019: Personal-Data Discovery and Privacy Evidence Inventory. Official problem statement.

These sources inform design choices; DATAQUEST is not a legal-compliance certification tool.

Thank You for Your Attention!

TraceLens · A3-PS019-TC279

Discover personal data. Trace its footprint. Explain the evidence.